



Anterior Abdominal Wall

Anatomy GIT Block · MEDucation by Mattin Majid

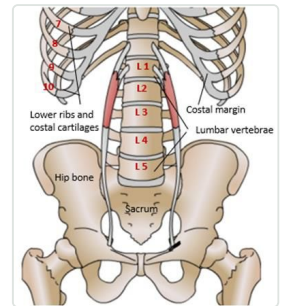
BEFORE YOU READ — ANSWER FROM MEMORY

1. Name four structures that lie on the transpyloric plane, and give its vertebral level.
2. Why does urine from a ruptured penile urethra never track into the thigh?
3. What forms the posterior wall of the rectus sheath below the arcuate line?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Abdomen — boundaries and surface landmarks

Abdomen	Lower trunk below the diaphragm, enclosing the abdominal cavity · no physical exam is complete without it
Boundaries	Above: diaphragm · below: continuous with pelvic cavity at the pelvic inlet · front: anterior wall · back: posterior wall
Costal margin	Cartilages of ribs 7–10
Xiphoid process	Apex of the subcostal angle · T9
Iliac crests	Highest points at L4 , just below the umbilicus (intercristal plane)*
Pubic symphysis	Lower limit of the anterior wall in the midline · level of the coccyx
Walls divided by	Midaxillary line separates anterior and posterior walls · bones: lower ribs and cartilages, lumbar vertebrae, sacrum, coccyx, hip bones

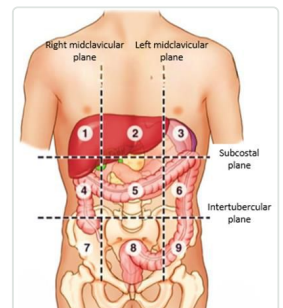


Bones and landmarks of the abdomen.

*The slide gives the crest summit at L4 but the plane “through L3–4”.

2 Planes and the nine regions

Plane	Where
Transpyloric (Addison's)	Midway between jugular notch and pubic symphysis · front: tip of 9th costal cartilage · back: lower border of L1
Subcostal	Lower border of the 10th costal cartilage → L3 · upper plane of the nine regions
Transumbilical	Through the umbilicus · L3/L4 disc
Intertubercular	Tubercles of the iliac crests · L5 · lower plane of the nine regions
Midclavicular (vertical)	Midpoint of clavicle → midway between ASIS and pubic symphysis



Nine regions formed by the subcostal, intertubercular and midclavicular planes.

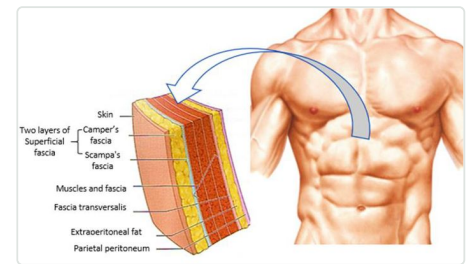
On the transpyloric plane (L1): pylorus · neck of pancreas · duodenojejunal flexure · fundus of gallbladder · hila of the kidneys · origins of the SMA and portal vein · tip of the 9th costal cartilage · end of the spinal cord (conus medullaris).

Region	Contents (per slides)
1 Right hypochondrium	Right lobe of liver, gallbladder, right colic flexure, right suprarenal, upper right kidney
2 Epigastrium	Pyloric stomach, liver, pancreas, duodenum, abdominal aorta
3 Left hypochondrium	Spleen, left kidney, left suprarenal, stomach, pancreas, splenic flexure
4 Right lumbar	Gallbladder, right kidney, part of liver, ascending colon
5 Umbilical	Omentum, mesentery, transverse colon, duodenum, jejunum, ileum
6 Left lumbar	Descending colon, left kidney, parts of jejunum and ileum, part of spleen
7 Right iliac	Cecum, appendix , right ureter, right ovary

8 Hypogastric (pubic)	Ileum, bladder
9 Left iliac	Descending colon, sigmoid colon , left ureter, left ovary

3 Anterior abdominal wall — six layers

#	Layer (superficial → deep)
1	Skin
2	Superficial fascia (Camper's + Scarpa's below the umbilicus)
3	Muscles (three flat + two vertical)
4	Fascia transversalis
5	Extraperitoneal tissue (fat)
6	Parietal peritoneum



Layers of the anterior abdominal wall.

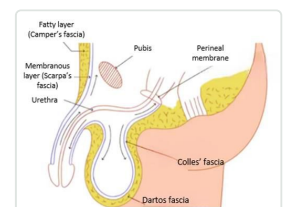
No deep fascia here, so the wall can distend after meals and in pregnancy (also absent in penis, scrotum, face and perineum). The wall matters because exams, incisions and most hernias go through it.

Skin and the umbilicus

Skin	Loosely attached except at the umbilicus · Langer's lines run horizontally → horizontal incisions heal with thinner scars
Umbilicus level	Adults L3-L4 · lower in newborns (less developed pelvis) · lower in old age (weaker muscle tone)
Watershed	Veins: above → lateral thoracic → axillary vein; below → superficial epigastric → great saphenous → femoral vein · Lymph: above → axillary nodes; below → superficial inguinal nodes
Other facts	Marks the T10 dermatome · a site of portocaval anastomosis · never incise across it (cosmetic)

Superficial fascia

	Camper's (fatty)	Scarpa's (membranous)
Nature	Fat, variable thickness	Thin, little or no fat
Into thigh	Continuous with superficial fascia of thigh	Attached to fascia lata → closed off from the thigh
Male	Loses fat in scrotum → dartos (smooth muscle)	Penis → Buck's fascia · perineum → Colles' fascia
Female	Fat in the labia majora	—



Camper's, Scarpa's, Colles' and dartos fascia.

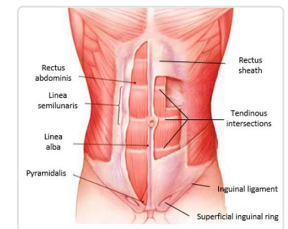
Between the layers: superficial epigastric, superficial circumflex iliac and external pudendal vessels, superficial inguinal nodes.

Ruptured penile urethra: urine collects **deep to Scarpa's fascia** → scrotum, perineum, penis and lower anterior abdominal wall, but **never the thigh**, because Scarpa's fascia is attached to the fascia lata.

4 Muscles

Muscle	Origin	Insertion	Nerve	Action
External oblique	External surfaces of ribs 5-12	Linea alba, pubic tubercle, iliac crest	Thoraco-abdominal (T7-T12)	Flexes and rotates trunk
Internal oblique	Thoracolumbar fascia, lateral 1/3 inguinal ligament	Ribs 10-12, linea alba, pecten pubis via conjoint tendon	Thoraco-abdominal + L1	Flexes and rotates trunk
Transversus abdominis	Thoracolumbar fascia, iliac crest, lateral 1/3 inguinal ligament	Linea alba, pecten pubis via conjoint tendon	Thoraco-abdominal + L1	Supports viscera
Rectus abdominis	Pubic symphysis and crest	Xiphoid, costal cartilages 5-7	Thoraco-abdominal	Flexes trunk

External oblique	Most superficial · fibres down, forward, medially (“hands in side pockets”) · lower free border rolls in → inguinal ligament (ASIS → pubic tubercle) · gap above pubis = superficial inguinal ring · posterior border = front of the lumbar triangle
Internal oblique	Fibres up, forward, medially · arches over the inguinal canal and fuses with transversus → conjoint tendon (pubic crest + medial pecten) · lower fibres → cremaster muscle and fascia
Transversus	Innermost flat muscle; fibres horizontal
Rectus abdominis	Beside the linea alba · lateral border = linea semilunaris · 3 tendinous intersections : xiphoid tip, umbilicus, midway between (sometimes a 4th below the umbilicus)
Pyramidalis	Small, rudimentary, in front of lower rectus inside the sheath · absent in ~20%
Functions	Support viscera · raise intra-abdominal pressure (defecation, micturition, delivery) · move trunk, maintain posture

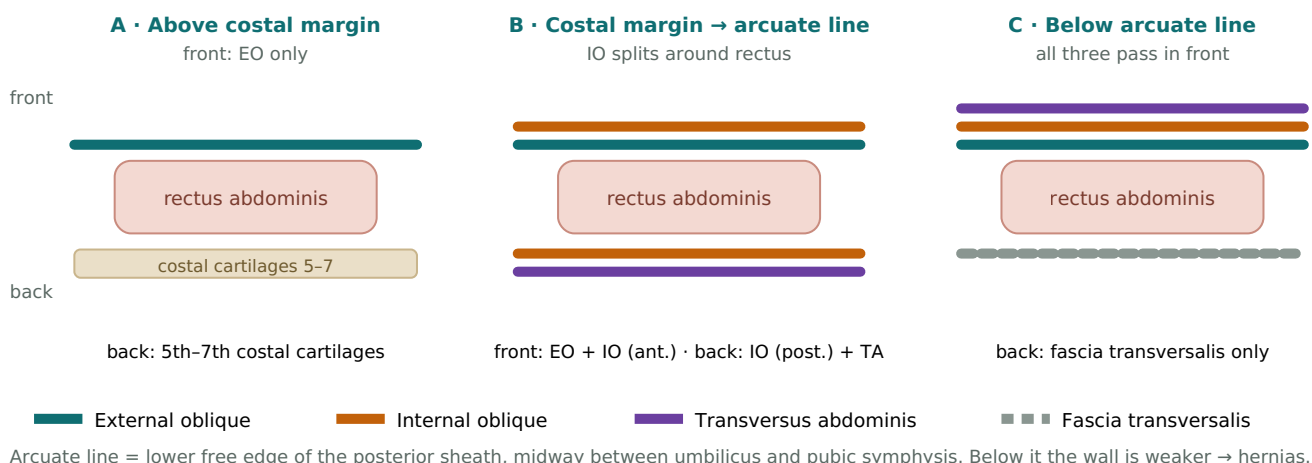


Rectus abdominis, linea alba, linea semilunaris and tendinous intersections.

5 Rectus sheath and deep layers

Aponeurotic sheath from the three flat muscles, enclosing **two muscles** (rectus, pyramidalis), **two nerves** and **two vessels** (per slides).

RECTUS SHEATH AT THREE LEVELS — which aponeurosis lies in front of and behind rectus



Fascia transversalis	Lines the inner surface of transversus · deep inguinal ring : oval opening ~ 1.2 cm above the mid-inguinal point (spermatic cord / round ligament) · its tube around the cord = internal spermatic fascia
Extraperitoneal fascia	Variable fat between fascia transversalis and peritoneum; continues into the pelvis
Parietal peritoneum	Thin serous membrane lining the abdominal walls

P Must-remember pearls

- Xiphoid **T9** · costal margin = cartilages 7–10 · iliac crest summit **L4** · umbilicus **L3/4**, T10 dermatome.
- Transpyloric plane = **L1**, tip of 9th costal cartilage: pylorus, pancreatic neck, DJ flexure, gallbladder fundus, renal hila, SMA/portal vein origins, conus.
- Nine regions: subcostal (L3) + intertubercular (L5) planes and two midclavicular lines.
- Six layers; **no deep fascia** in the anterior abdominal wall.
- Umbilical watershed: above → **axillary** veins and nodes; below → **femoral** vein and **superficial inguinal** nodes.

- Camper's → dartos in scrotum · Scarpa's → Buck's (penis), Colles' (perineum), attached to **fascia lata**.
- Inguinal ligament = rolled lower border of external oblique (ASIS → pubic tubercle) · conjoint tendon = IO + TA · cremaster from IO.
- Rectus: 3 tendinous intersections (xiphoid, umbilicus, between) · linea semilunaris = lateral border · pyramidalis absent in 20%.
- Below the arcuate line all three aponeuroses go in front; behind is **fascia transversalis only** → hernia-prone.
- Deep inguinal ring: fascia transversalis, 1.2 cm above the mid-inguinal point.

Q Test yourself

1. A 60-year-old man has left iliac fossa pain with constipation and mucus in his stools. Colonoscopy shows diverticulosis. Which organ is most likely the source?

- | | |
|----------------|--------------------|
| A. Pancreas | B. Sigmoid colon |
| C. Appendix | D. Small intestine |
| E. Gallbladder | |

2. After a straddle injury a man's penile urethra ruptures. Urine collects in the scrotum and lower abdominal wall but not the thigh. Which attachment explains this?

- | | |
|---|--|
| A. Camper's fascia to the inguinal ligament | B. Fascia transversalis to the iliac crest |
| C. External oblique to the pubic tubercle | D. Scarpa's fascia to the fascia lata |
| E. Dartos fascia to the perineal body | |

3. During hernia repair the surgeon sees the posterior rectus sheath end in a curved free margin. Below it, what forms the posterior wall of the sheath?

- | | |
|--------------------------------------|---|
| A. Transversus abdominis aponeurosis | B. Posterior lamina of internal oblique |
| C. Fascia transversalis only | D. Costal cartilages 5–7 |
| E. External oblique aponeurosis | |

4. A CT slice through the lower border of L1 is reviewed. Which structure is expected on this plane?

- | | |
|------------------------------|-----------------------------|
| A. Neck of the pancreas | B. Sigmoid colon |
| C. Umbilicus | D. Bifurcation of the aorta |
| E. Superficial inguinal ring | |

5. Lymph from skin just below the umbilicus drains first to which nodes?

- | | |
|-------------------------|-------------------|
| A. Axillary | B. Deep cervical |
| C. Para-aortic | D. Internal iliac |
| E. Superficial inguinal | |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 B — The sigmoid colon lies in the left iliac region and is the commonest site of diverticula.

2 D — Scarpa's (membranous) fascia is attached to the fascia lata, closing the space off from the thigh.

3 C — Below the arcuate line all three aponeuroses pass in front of rectus; behind it is only fascia transversalis.

4 A — The transpyloric plane (lower border of L1) passes through the pancreatic neck, pylorus, DJ flexure, gallbladder fundus and renal hila.

5 E — The umbilicus is a lymphatic watershed: below it lymph drains to the superficial inguinal nodes, above it to the axillary nodes.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy SGL 1 slides, 2026-2027

Study alongside the lecture, not instead of it.